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A Directed-Hypergraph Representation of Higher-Order Information Flows

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A statistical approach to higher-order interactions requires an understanding of correlations among components. Their temporal evolution can be described using multiple transfer entropy and multiple reversed transfer entropy. This paper investigates the relationship between the topology of these information flows and the entropy dynamics of individual components. We introduce an information flow hypergraph using forward dynamical neighbors for transfer entropy and backward dynamical neighbors for reversed transfer entropy. Under deterministic reversible dynamics, the two neighbor sets provide an exact local entropy balance without a residual term. An additional strong blocking flows condition matches reversed-transfer-entropy hyperarcs to the spatial reversals of transfer-entropy hyperarcs. This correspondence relates B-inflows and F-outflows to the two nonnegative terms in the entropy balance. A BF-sink has strictly increasing entropy, whereas a BF-source has strictly decreasing entropy. For a fixed nonempty flow topology, periodic or recurrent entropy dynamics therefore requires a weak BF-hypercycle. Stationarity further requires both a B-inflow and an F-outflow at every non-isolated vertex, together with numerical balance of their aggregate contributions.

Keywords: mutual information; transfer entropy; information dynamics; higher-order correlation; complex network; directed hypergraphs

1. Introduction

Synergistic effects are key characteristics of higher-order dynamics [1–7] and distinguish many-body interactions from purely pairwise interactions. Several studies [8–11] use information theory to characterize such dynamics. In particular, partial information decomposition (PID) [11, 12] provides nonnegative measures of synergistic and redundant information, although their definitions are not unique. Here, we instead use multiple mutual information, also called interaction information or co-information [8, 13–19]. This signed quantity has a standard inclusion–exclusion definition at every order and describes higher-order correlation structures beyond ordinary pairwise mutual information [8, 9].

The temporal evolution of these correlations can be decomposed using multiple transfer entropy and multiple reversed transfer entropy [20]. These quantities extend the decomposition of mutual-information dynamics to higher orders. Each has a source set and a target set, which naturally correspond to the tail and head of a hyperarc [21–23]. However, nonzero transfer entropy may reflect indirect statistical dependencies. We therefore distinguish direct information flows using dynamical-neighbor sets. Forward neighbors determine a component's future state, whereas backward neighbors reconstruct its past state from future variables. This distinction treats the two temporal directions consistently.

The resulting information flow hypergraph contains a BF-hypergraph substructure composed of B-arcs and F-arcs [21, 22, 24]. Under deterministic reversible dynamics, forward and backward neighbor

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sets give an exact entropy balance for each vertex. We separately impose a strong blocking flows condition that matches the support of reversed transfer entropy to the spatial reversal of the transfer-entropy support. This is an additional restriction on the dynamics, rather than a consequence of stationarity. A closed copying cycle illustrates a system satisfying both reversibility and strong blocking, while an autonomous control example shows that reversibility alone does not imply this support condition.

The support correspondence makes the entropy at a BF-sink strictly increase and the entropy at a BF-source strictly decrease. These conclusions extend the pairwise argument in Ref. [25]. A finite BF-hypergraph without weak BF-hypercycles has both types of vertex. Consequently, when the flow topology is fixed and the assumptions hold throughout the evolution, periodic or recurrent entropy dynamics requires a weak BF-hypercycle. Cycle existence is not automatic under the assumptions: a reversible swap can transfer information along a single directed edge.

A weak BF-hypercycle alone does not establish entropy stationarity. We derive an additional necessary local condition: every non-isolated vertex must have both a B-inflow and an F-outflow. Their existence does not enforce numerical equality of the two aggregate entropy contributions. An ordinary directed cycle satisfies this topological condition [26, 27], and the copying cycle provides an explicit stationary realization.

Section 2 reviews the information-theoretic and hypergraph definitions. Section 3 introduces the forward and backward flow hypergraphs, the exact entropy balance, and the blocking conditions. Section 4 analyzes BF-sinks, BF-sources, and cycle necessity. Section 5 derives the local condition for entropy stationarity. The subsequent analysis concerns individual entropy dynamics; it does not claim that the selected hyperarcs reconstruct every higher-order mutual-information balance.

2. Definitions

2.1 *Higher-Order Information Dynamics*

Consider an N-component system whose components are indexed by a finite set V. The state of $v \in V$ at time t is denoted by $\Gamma_v(t)$ and takes values in a discrete state space. An ensemble of states induces a joint probability distribution. All entropies used below are assumed finite. Unprimed and primed variables refer to times t and $t+\Delta t$, respectively, where $\Delta t > 0$. For a vertex set A, A' denotes the corresponding collection of future variables. Logarithms are natural, so information is measured in nats. The symbol $\subseteq$ includes equality, and sums over nonempty subsets are zero when the indexing set is empty.

DEFINITION 2.1 For $v \in V$, the **Shannon entropy** [28, 29] is

$$H_v := -\sum_{\gamma_v} \mathbb{P}(\Gamma_v = \gamma_v) \log \mathbb{P}(\Gamma_v = \gamma_v). \tag{2.1}$$

For a nonempty set $A = \{a_1, \ldots, a_k\}$, its joint Shannon entropy is

$$H_A := -\sum_{\gamma_{a_1}, \ldots, \gamma_{a_k}} \mathbb{P}(\Gamma_A = \gamma_A) \log \mathbb{P}(\Gamma_A = \gamma_A). \tag{2.2}$$

We set $H_\varnothing = 0$. The **conditional entropy** is

$$H_B|_A := H_{A\cup B} - H_A. \tag{2.3}$$

DEFINITION 2.2 The **mutual information** and **conditional mutual information** [29] are

$$I(A;B) := H_A + H_B - H_{A\cup B}, \qquad I(A;B \mid C) := H_A|_C + H_B|_C - H_{A\cup B}|_C\,. \tag{2.4}$$

Each set in these expressions is treated as a single joint random variable. For singleton sets, we also write $I_{v:u} = I(v;u)$.

For discrete variables, Shannon entropy, conditional entropy, and conditional mutual information are nonnegative.

DEFINITION 2.3 For a nonempty collection A of variables, the **multiple mutual information (co-information)** [13, 14, 16] is

$$I_A^{\mathrm{co}} := \sum_{r=1}^{|A|} (-1)^{r+1} \sum_{\substack{S\subseteq A \\ |S|=r}} H_S. \tag{2.5}$$

Its conditional version is

$$I_A^{\mathrm{co}}\,|_B := \sum_{r=1}^{|A|} (-1)^{r+1} \sum_{\substack{S\subseteq A \\ |S|=r}} H_S\,|_B\,. \tag{2.6}$$

For $|A| = 2$, co-information reduces to mutual information; for $|A| = 1$, it reduces to Shannon entropy. At higher orders it may be positive, negative, or zero. Unlike $I(A;B \mid C)$, co-information treats the members of its argument set separately.

The temporal evolution of correlations can be related to directional information flows [20]. Transfer entropy provides a statistical measure of directional predictive dependence [30].

DEFINITION 2.4 The one-step **transfer entropy** [31, 32] from u to v is

$$T_{u\to v} := I(v';u \mid v). \tag{2.7}$$

DEFINITION 2.5 **Reversed transfer entropy** [33–35], also called backward transfer entropy [33], reverses the temporal indexing:

$$T^{\dagger}_{u\to v} := I(v;u' \mid v'). \tag{2.8}$$

DEFINITION 2.6 For nonempty disjoint vertex sets $A, B \subseteq V$, **multiple transfer entropy** and **multiple reversed transfer entropy** [20] are

$$T_{B\to A} := I^{\mathrm{co}}_{A'\cup B}\,|_A, \qquad T^{\dagger}_{B\to A} := I^{\mathrm{co}}_{A\cup B'}\,|_{A'}\,. \tag{2.9}$$

These expressions reduce to ordinary transfer entropy and reversed transfer entropy when $|A| = |B| = 1$.

For any $L \subseteq V \setminus \{v\}$, inclusion–exclusion gives the exact entropy identity [20]

$$\Delta H_v = \sum_{\varnothing \neq S\subseteq L} (-1)^{|S|+1} \left(T_{S\to\{v\}} - T^{\dagger}_{S\to\{v\}}\right) + \alpha_1^{L\to v}, \tag{2.10}$$

where $\Delta H_v := H_{v'} - H_v$ and

$$\alpha_1^{L\to v} := H_{v'}\,|_{\{v\}\cup L} - H_v\,|_{\{v'\}\cup L'}\,. \tag{2.11}$$

The residual α_1 contains contributions not captured by the displayed flow terms. Section 3.2 distinguishes this arbitrary expansion set from forward-minimal and backward-minimal neighbor sets.

2.2 *Directed Hypergraphs*

We summarize the definitions needed below [21–23, 36].

DEFINITION 2.7 A **directed hypergraph** $\mathscr{G} = (V, \overrightarrow{E})$ consists of a finite vertex set and a nonempty set of ordered pairs (e^+, e^-), where $e^+, e^- \subseteq V$ are nonempty and disjoint. The pair is a **hyperarc**, with tail $\mathscr{T}(e) = e^+$ and head $\mathscr{H}(e) = e^-$. Define

$$\overleftarrow{E} := \{(e^-, e^+) : (e^+, e^-) \in \overrightarrow{E}\}. \tag{2.12}$$

The hypergraph $(V, \overleftarrow{E})$ is the **symmetric image** of $\mathscr{G}$.

Unless stated otherwise, the hypergraphs to which the graph-theoretic results are applied have no isolated vertices:

$$V = \bigcup_{e \in \overrightarrow{E}} \big[\mathscr{T}(e) \cup \mathscr{H}(e)\big]. \tag{2.13}$$

DEFINITION 2.8 A directed hypergraph is **oriented** if $\overrightarrow{E} \cap \overleftarrow{E} = \varnothing$.

DEFINITION 2.9 A **BF-hypergraph** is a directed hypergraph in which every hyperarc satisfies $|e^+| = 1$ or $|e^-| = 1$. A hyperarc with $|e^-| = 1$ is a **B-arc**; one with $|e^+| = 1$ is an **F-arc**. An ordinary directed edge is both a B-arc and an F-arc.

DEFINITION 2.10 For a vertex v, define

$$\begin{aligned} &\mathrm{In}_B(v) := \{(e^+, e^-) \in \overrightarrow{E} : e^- = \{v\}\}, && \mathrm{Out}_B(v) := \{(e^+, e^-) \in \overrightarrow{E} : v \in e^+,\ |e^-| = 1\}, \\ &\mathrm{In}_F(v) := \{(e^+, e^-) \in \overrightarrow{E} : v \in e^-,\ |e^+| = 1\}, && \mathrm{Out}_F(v) := \{(e^+, e^-) \in \overrightarrow{E} : e^+ = \{v\}\}. \end{aligned} \tag{2.14}$$

These are the B-inflow, B-outflow, F-inflow, and F-outflow sets, respectively. The same definitions apply to the BF-hypergraph substructure of a general directed hypergraph.

DEFINITION 2.11 A **BF-sink** satisfies $\mathrm{Out}_B(v) = \mathrm{Out}_F(v) = \varnothing$ and $\mathrm{In}_B(v) \cup \mathrm{In}_F(v) \neq \varnothing$. A **BF-source** satisfies $\mathrm{In}_B(v) = \mathrm{In}_F(v) = \varnothing$ and $\mathrm{Out}_B(v) \cup \mathrm{Out}_F(v) \neq \varnothing$.

DEFINITION 2.12 A **weak hyperpath** from v_1 to v_{n+1} is a sequence $(v_1, e_1, v_2, e_2, \ldots, e_n, v_{n+1})$ such that $v_i \in \mathscr{T}(e_i)$ and $v_{i+1} \in \mathscr{H}(e_i)$ for $1 \leqslant i \leqslant n$. A vertex s is **connected to** u if a weak hyperpath leads from u to s. A nonempty set S is connected to a nonempty set U if some $u \in U$ has a weak hyperpath to some $s \in S$.

DEFINITION 2.13 The **transitive closure** $\mathscr{G}^* = (V, \overrightarrow{E}^*)$ contains (U, S), for nonempty disjoint U, S, if and only if S is connected to U in $\mathscr{G}$.

DEFINITION 2.14 A **weak BF-hypercycle** is a weak hyperpath in a BF-hypergraph with $v_1 = v_{n+1}$ and $n \geqslant 1$. No distinctness condition is imposed on the intermediate vertices or hyperarcs.

3. Identification of Higher-Order Information Flows

3.1 *Forward and Backward Information Flow Hypergraphs*

Let V be the component set. The total support of multiple transfer entropy is

$$\overrightarrow{E}_{\text{total}} := \{(A, B) : A, B \subseteq V \text{ are nonempty and disjoint, } T_{A \to B} \neq 0\}. \tag{3.1}$$

It can contain indirect statistical relationships. To select flows associated with dynamical dependencies, we distinguish the two temporal directions.

DEFINITION 3.1 A set of **forward dynamical neighbors** $K_A \subseteq V \setminus A$ is an inclusion-minimal set satisfying

$$H_{A'}|_{A\cup K_A} = 0. \tag{3.2}$$

A set of **backward dynamical neighbors** $Q_A \subseteq V \setminus A$ is an inclusion-minimal set satisfying

$$H_A|_{A'\cup Q'_A} = 0. \tag{3.3}$$

We write $K_v := K_{\{v\}}$ and $Q_v := Q_{\{v\}}$.

The members of Q_A are vertex labels; the conditioning variables in Eq. (3.3) are their future states. Forward and backward neighbor sets need not coincide and need not be unique. Let

$$\begin{aligned} \overline{\mathscr{K}}_A &:= \{K : K \text{ is a forward dynamical-neighbor set of } A\}, \\ \overline{\mathscr{Q}}_A &:= \{Q : Q \text{ is a backward dynamical-neighbor set of } A\}. \end{aligned} \tag{3.4}$$

We take the state dynamics to be deterministic and reversible on the ensemble support:

$$\Gamma_V(t+\Delta t) = F_t(\Gamma_V(t)), \qquad \Gamma_V(t) = \widetilde{F}_t(\Gamma_V(t+\Delta t)). \tag{3.5}$$

Thus $V \setminus A$ satisfies each sufficiency condition, and inclusion-minimal subsets exist because V is finite. The distributions may be nonstationary, and F_t may depend on time. Reversibility here concerns reconstruction of the full state; it does not require $K_A = Q_A$.

The direct forward support is

$$\overrightarrow{E}_T := \{(A,B) \in \overrightarrow{E}_{\text{total}} : \exists K_B \in \overline{\mathscr{K}}_B \text{ such that } A \subseteq K_B\}. \tag{3.6}$$

Directness is defined relative to the ensemble and the chosen component representation. Since minimal sufficient sets can be nonunique, this definition admits every qualifying choice.

PROPOSITION 3.2 In general, $\mathscr{G}_T^* \neq \mathscr{G}_{\text{total}}^*$, where $\mathscr{G}_T = (V, \overrightarrow{E}_T)$ and $\mathscr{G}_{\text{total}} = (V, \overrightarrow{E}_{\text{total}})$.

Proof. A deterministic reversible counterexample is given in Appendix A. □

For comparison, consider the neighbor-anchored support

$$\overrightarrow{E}_{\bar{T}} := \{(A,B) \in \overrightarrow{E}_{\text{total}} : \exists K_B \in \overline{\mathscr{K}}_B \text{ such that } A \cap K_B \neq \varnothing\}. \tag{3.7}$$

In the counterexample of Appendix A, $T_{\{Y,C\}\to\{X\}} = \ln 2 - h(\varepsilon) > 0$. The enlarged source set intersects $K_X = \{C\}$, so the corresponding hyperarc connects Y to X in $\mathscr{G}_{\bar{T}}$. More generally, this modification preserves total connectivity.

PROPOSITION 3.3 $\mathscr{G}_{\bar{T}}^* = \mathscr{G}_{\text{total}}^*$, where $\mathscr{G}_{\bar{T}} = (V, \overrightarrow{E}_{\bar{T}})$.

Proof. See Appendix B. □

The subsequent analysis uses $\overrightarrow{E}_T$, whose source sets are contained in a forward dynamical-neighbor set, rather than the larger neighbor-anchored support.

DEFINITION 3.4 The **transfer-entropy hypergraph** is $\mathscr{G}_T = (V, \overrightarrow{E}_T)$, with $\overrightarrow{E}_T$ given by Eq. (3.6). The **reversed-transfer-entropy hypergraph** is $\mathscr{G}_{T^\dagger} = (V, \overrightarrow{E}_{T^\dagger})$, where

$$\overrightarrow{E}_{T^\dagger} := \left\{(A,B) : A,B \subseteq V \text{ are nonempty and disjoint, } T^\dagger_{A\to B} \neq 0,\ \exists Q_B \in \overline{\mathscr{Q}}_B \text{ with } A \subseteq Q_B\right\}. \tag{3.8}$$

The reversed hypergraph uses backward reconstruction to select direct flows. Consequently, every nonzero term in the backward-neighbor entropy expansion below is represented. The two definitions are temporal counterparts, although their supports need not be spatial reversals of one another.

DEFINITION 3.5 The **information flow hypergraph** is the ordered pair $\mathscr{G}_{\Delta I} := (\mathscr{G}_T, \mathscr{G}_{T^\dagger})$.

3.2 *Forward and Backward Reconstruction of Entropy*

Fix $K_v \in \overline{\mathscr{K}}_v$ and $Q_v \in \overline{\mathscr{Q}}_v$, and set

$$L_v := K_v \cup Q_v. \tag{3.9}$$

The combined set need not be inclusion-minimal. It is used for entropy bookkeeping, not to redefine forward directness. By Eqs. (3.2) and (3.3),

$$\alpha_1^{L_v\to v} = H_{v'}|_{\{v\}\cup L_v} - H_v|_{\{v'\}\cup L'_v} = 0. \tag{3.10}$$

This conclusion does not require $\alpha_1^{K_v\to v} = 0$ for the smaller forward-neighbor set. The distinction is essential when information about v is recovered from future variables outside K_v.

It is more convenient for the graph analysis to expand the two temporal directions separately:

$$\begin{aligned}\Delta H_v &= I(K_v; v' \mid v) - I(Q'_v; v \mid v') \\ &= \sum_{\varnothing\neq S\subseteq K_v} (-1)^{|S|+1} T_{S\to\{v\}} - \sum_{\varnothing\neq R\subseteq Q_v} (-1)^{|R|+1} T^\dagger_{R\to\{v\}}.\end{aligned} \tag{3.11}$$

Indeed, the sufficiency conditions give

$$f_v := I(K_v; v' \mid v) = H_{v'}|_v, \qquad r_v := I(Q'_v; v \mid v') = H_v|_{v'}, \qquad \Delta H_v = f_v - r_v. \tag{3.12}$$

Both f_v and r_v are nonnegative and independent of the particular admissible neighbor sets. Every nonzero flow term in Eq. (3.11) belongs to its corresponding hypergraph by definition. This local completeness does not assert completeness for all higher-order mutual-information balances.

3.3 *Blocking Flows Conditions*

To exclude a two-step weak cycle formed by a hyperarc and its spatial reverse, we impose a support restriction [20].

DEFINITION 3.6 The **blocking flows condition** is

$$\overrightarrow{E}_T \cap \overleftarrow{E}_T = \varnothing, \qquad \overrightarrow{E}_{T^\dagger} \cap \overleftarrow{E}_{T^\dagger} = \varnothing, \qquad \overrightarrow{E}_T \cap \overrightarrow{E}_{T^\dagger} = \varnothing. \tag{3.13}$$

Thus each flow hypergraph is oriented, and the same directed hyperarc cannot carry both represented quantities. These conditions constrain the chosen model and distribution. Replacing a bidirectional interaction by separate channels can be a modeling choice, but is not an automatic transformation preserving a given one-step stochastic process.

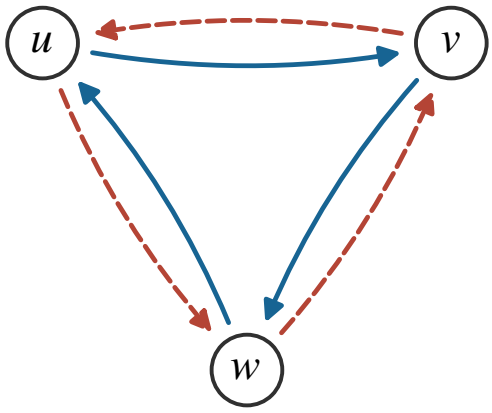


FIG. 1. The reversible copying cycle in Eq. (3.15). Solid blue arrows show TE and dashed red arrows show reversed TE. Every represented quantity equals $\ln 2$.

DEFINITION 3.7 The **strong blocking flows condition** is the blocking flows condition together with

$$\overrightarrow{E}_{T^\dagger} = \overleftarrow{E}_T. \tag{3.14}$$

This equality matches supports, not numerical flow values. It is an additional assumption and does not follow from a stationary aggregate balance. Under it, $\mathcal{G}_{T^\dagger}$ is the symmetric image of $\mathcal{G}_T$. A matched pair $(A,B) \in \overrightarrow{E}_T$ and $(B,A) \in \overrightarrow{E}_{T^\dagger}$ requires $A \subseteq K_B$ for some forward-neighbor set and $B \subseteq Q_A$ for some backward-neighbor set. It does not require mutual forward causal parenthood.

A closed copying cycle gives a concrete realization. Let u,v,w be independent fair bits and let

$$v' = u, \qquad u' = w, \qquad w' = v. \tag{3.15}$$

This map is a permutation, and the independent uniform distribution is stationary. At v,

$$K_v = \{u\}, \qquad Q_v = \{w\}, \qquad L_v = \{u,w\}, \tag{3.16}$$

and the other neighbor sets follow cyclically. The only nonzero TE and reversed-TE hyperarcs are

$$\begin{aligned} \overrightarrow{E}_T &= \{(\{u\},\{v\}),(\{v\},\{w\}),(\{w\},\{u\})\}, \\ \overrightarrow{E}_{T^\dagger} &= \overleftarrow{E}_T, \end{aligned} \tag{3.17}$$

with value $\ln 2$ on every represented arc. All higher-order terms vanish. Hence strong blocking and $\alpha_1^{L_x \to x} = 0$ hold at every vertex x. In contrast, $\alpha_1^{K_v \to v} = -\ln 2$ for the smaller forward set: v cannot be reconstructed from $v' = u$ and $u' = w$. Figure 1 shows the support correspondence.

A six-node extension realizes higher-order channels within the same framework. Let a,b,c,d,e,f be independent fair bits and define

$$a' = e, \qquad b' = f, \qquad c' = a, \qquad d' = a \oplus b, \qquad e' = c, \qquad f' = d. \tag{3.18}$$

This is a reversible XOR stage routed through a three-layer cycle. Its inverse is

$$a = c', \qquad b = c' \oplus d', \qquad c = e', \qquad d = f', \qquad e = a', \qquad f = b'. \tag{3.19}$$

The independent uniform distribution is invariant. The direct TE support contains five copying edges, $a \to c$, $c \to e$, $d \to f$, $e \to a$, and $f \to b$, each with value $\ln 2$, and the three higher-order channels

$$T_{\{a,b\}\to\{d\}} = T_{\{b\}\to\{c,d\}} = T_{\{a,b\}\to\{c,d\}} = -\ln 2. \tag{3.20}$$

All other direct TE terms vanish. The reversed support is exactly the spatial reversal of this list, with matching values, so strong blocking holds. For example, $K_b = \{f\}$ and $Q_b = \{c,d\}$, whereas $K_d = \{a,b\}$ and $Q_d = \{f\}$. Thus both temporal directions are covered, and every combined-set residual vanishes. This example realizes B-, F-, and multiple-tail/multiple-head channels under all assumptions of the main results.

The condition is restrictive even among reversible systems. For independent fair bits, the controlled update

$$u' = u, \qquad v' = v \oplus u \tag{3.21}$$

is reversible, with $K_v = Q_v = \{u\}$. However, $T_{u\to v} = T^{\dagger}_{u\to v} = \ln 2$, so ordinary blocking already fails. More generally, a copying channel $v' = u$ has $T^{\dagger}_{v\to u} = H_u|_{u'}$; if u' alone reconstructs u, the reverse flow vanishes and cannot match a nonzero copying TE.

Strong blocking also permits a source and a sink without a cycle. Initially let v be a fair bit and $u = 0$, and apply

$$v' = u, \qquad u' = v. \tag{3.22}$$

Then $K_v = \varnothing$, $Q_v = \{u\}$, $K_u = \{v\}$, and $Q_u = \varnothing$. Both combined-set residuals vanish. The support is the single edge $v \to u$, its reversed-TE counterpart is $u \to v$, and both values are $\ln 2$. Thus v is a BF-source and u a BF-sink, with $\Delta H_v = -\ln 2$ and $\Delta H_u = \ln 2$. Repeating the swap reverses the instantaneous support at each step. This example explains why a cycle-necessity statement for recurrent dynamics must specify its temporal topology assumption. Appendix C separately illustrates local higher-order channels driven by stochastic inputs.

4. BF-Sinks, BF-Sources, and Cycle Necessity

Let $\overrightarrow{E}_{BF} \subseteq \overrightarrow{E}_T$ contain all B- and F-arcs, and let V_{BF} be their incident vertices. These define the BF-hypergraph substructure $\mathscr{G}_{BF} = (V_{BF}, \overrightarrow{E}_{BF})$. Under the reversible dynamics and strong blocking used below, Proposition 4.2 shows that every non-isolated vertex of $\mathscr{G}_T$ belongs to this substructure.

PROPOSITION 4.1 A finite BF-hypergraph without isolated vertices or weak BF-hypercycles has a BF-sink and a BF-source.

Proof. If no BF-sink exists, every vertex has an outgoing B- or F-arc. Starting at any vertex, choose such an arc and then a vertex in its head, and repeat. Since the vertex set is finite, a vertex eventually repeats, producing a weak BF-hypercycle. This contradicts the assumption. Applying the same argument to the symmetric image proves the existence of a BF-source. □

LEMMA 4.1 For any $L \subseteq V \setminus \{v\}$,

$$I(L; v' \mid v) = 0 \quad \Longleftrightarrow \quad T_{S\to\{v\}} = 0 \text{ for every nonempty } S \subseteq L. \tag{4.1}$$

The temporally reversed statement holds with $I(L'; v \mid v')$ and $T^{\dagger}_{S\to\{v\}}$.

Proof. Inclusion–exclusion gives

$$I(L; v' \mid v) = \sum_{\varnothing \neq S \subseteq L} (-1)^{|S|+1} T_{S \to \{v\}}, \tag{4.2}$$

so vanishing of all terms implies vanishing of the aggregate. Conversely, the chain rule and nonnegativity give $I(S; v' \mid v) = 0$ for every $S \subseteq L$ when the aggregate is zero. For singleton S, the corresponding TE is zero. Inducting on $|S|$ in Eq. (4.2), applied to S, then shows that each higher-order term is zero. Reversing the temporal indexing proves the second statement. □

PROPOSITION 4.2 Under deterministic reversible dynamics and strong blocking,

$$\mathrm{In}_B(v) \neq \varnothing \iff f_v > 0, \qquad \mathrm{Out}_F(v) \neq \varnothing \iff r_v > 0. \tag{4.3}$$

Every non-isolated vertex has a B-inflow or an F-outflow. In addition, an F-inflow implies a B-inflow, and a B-outflow implies an F-outflow.

Proof. If $\mathrm{In}_B(v) = \varnothing$, all forward terms indexed by subsets of any $K_v \in \overline{\mathscr{K}}_v$ vanish. Lemma 4.1 gives $f_v = 0$. Conversely, if $f_v = H_{v'}|_v = 0$, then v' is determined by v, so every TE co-information with v in its target set vanishes. In particular, all B-inflows and F-inflows vanish. This proves the first equivalence and the F-inflow implication.

Strong blocking identifies an F-outflow at v with a B-inflow at v in $\mathscr{G}_{T^\dagger}$. Applying the reversed part of Lemma 4.1 to Q_v proves the second equivalence. More explicitly, absence of F-outflows eliminates all backward-neighbor terms, giving $r_v = 0$. If $r_v = H_v|_{v'} = 0$, every reversed-TE co-information whose target contains v vanishes. Strong blocking therefore excludes *every* outgoing TE hyperarc incident to v, including those with multiple vertices in both tail and head. This proves the B-outflow implication.

If both f_v and r_v vanish, the preceding arguments exclude all incoming and all outgoing TE hyperarcs, so v is isolated. The contrapositive proves the remaining assertion. □

The proposition does not require every non-isolated vertex to have a B-inflow. The swap in Eq. (3.22) is a counterexample to that stronger assertion. However, it ensures that every active vertex is incident to a B- or F-arc, so $V_{BF} = V$ under the no-isolated-vertices convention.

PROPOSITION 4.3 Under deterministic reversible dynamics and strong blocking, a BF-sink v has strictly increasing entropy: $\Delta H_v > 0$.

Proof. A BF-sink has no F-outflow, hence $r_v = 0$ by Proposition 4.2. It has a B-inflow or an F-inflow, and either gives a B-inflow by the same proposition. Thus $f_v > 0$, and Eq. (3.12) yields

$$\Delta H_v = f_v = I(K_v; v' \mid v) > 0. \tag{4.4}$$

□

COROLLARY 4.1 Under the same assumptions, a BF-source v has strictly decreasing entropy: $\Delta H_v < 0$.

Proof. A BF-source has no B-inflow, so $f_v = 0$. It has a B-outflow or an F-outflow. The former implies the latter by Proposition 4.2; hence $r_v > 0$. Therefore

$$\Delta H_v = -r_v = -I(Q'_v; v \mid v') < 0. \tag{4.5}$$

□

These conclusions are temporal counterparts. Cases with an F-inflow but no B-inflow are impossible: the absence of a B-inflow gives $H_{v'}|_v = 0$, and a co-information term vanishes whenever one of its

arguments is determined by the conditioning variables. Similarly, strong blocking excludes a B-outflow without an F-outflow.

COROLLARY 4.2 Suppose the nonempty TE support is fixed throughout the evolution, deterministic reversibility and strong blocking hold at every step, and the active vertex set is finite. Periodic or recurrent entropy dynamics requires a weak BF-hypercycle.

Proof. Without a weak BF-hypercycle, Proposition 4.1 gives a BF-sink and a BF-source in the fixed BF substructure. Their entropies change strictly in opposite directions at every step. In particular, after the first step each differs from its initial value by a strictly positive amount in a fixed direction, precluding a periodic return or arbitrarily close returns of the entropy vector to that value. □

Here recurrence means returns of the entropy vector arbitrarily close to a previous value at arbitrarily late times. Strict monotonicity does not exclude asymptotic relaxation. When the support changes with time, the corollary does not apply, as demonstrated by the repeated swap. Nor does the existence of a cycle establish that every vertex lies on a cycle: directed cycles can be connected by one-way paths whose intermediate vertices belong to no cycle.

5. Entropy Balance on Weak BF-Hypercycles

The previous section gives a necessary cycle condition under a fixed topology. The presence of a weak BF-hypercycle alone does not establish stationarity. The local distinction between B-inflow and F-outflow determines whether the two nonnegative contributions to ΔH_v can balance.

LEMMA 5.1 Under deterministic reversible dynamics and strong blocking, the following hold at a vertex v:

(a) If v has a B-inflow but no F-outflow, then $\Delta H_v > 0$.

(b) If v has an F-outflow but no B-inflow, then $\Delta H_v < 0$.

(c) If v has both a B-inflow and an F-outflow, then

$$\Delta H_v = f_v - r_v, \qquad f_v > 0, \quad r_v > 0. \tag{5.1}$$

The support information alone does not fix the sign of this difference.

Proof. The three cases follow from the support equivalences in Eq. (4.3) and the exact entropy balance in Eq. (3.12). □

If neither a B-inflow nor an F-outflow is present, both rates vanish; Proposition 4.2 then implies that the vertex is isolated. Thus this case is absent from the active hypergraph. The lemma does not reintroduce the impossible combinations involving an F-inflow without a B-inflow.

PROPOSITION 5.1 Under deterministic reversible dynamics and strong blocking, a necessary condition for stationarity of $(H_v)_{v\in V}$ is that every non-isolated vertex has both a B-inflow and an F-outflow. Numerical entropy stationarity at each vertex requires

$$I(K_v; v' \mid v) = I(Q'_v; v \mid v'). \tag{5.2}$$

Proof. Stationarity gives $f_v = r_v$. At a non-isolated vertex at least one rate is positive by Proposition 4.2, so both are positive. Equation (4.3) gives both required flow types. Equality (5.2) is exactly $\Delta H_v = 0$ by Eq. (3.12). □

This is a necessary topological condition, not a sufficient condition for stationarity of the joint distribution or of all information-theoretic quantities. Even the presence of both flow types does not enforce equality of their numerical contributions. Under a fixed topology, the condition also implies the existence of a weak BF-hypercycle in the finite active graph. It does not imply that every vertex lies on such a cycle.

An ordinary directed cycle satisfies the local topological condition because each directed edge is both a B-arc and an F-arc. The copying cycle in Eq. (3.15) supplies a stationary realization with $f_v = r_v = \ln 2$ at every vertex. A weak BF-hypercycle with a vertex lacking one of the two required flow types cannot support entropy stationarity under the stated assumptions.

6. Conclusion

We introduced an information flow hypergraph for forward and reversed multiple transfer entropy. Forward and backward dynamical neighbors define the two temporal directions separately. Their union makes the residual α_1 vanish, while separate expansions over the two neighbor sets provide complete local coverage of the entropy balance.

Strong blocking is an additional support-matching assumption. It relates B-inflows and F-outflows in the TE graph to the nonnegative quantities $H_{v'}|_v$ and $H_v|_{v'}$. A BF-sink therefore has strictly increasing entropy, whereas a BF-source has strictly decreasing entropy. When the nonempty support is fixed, a finite acyclic BF substructure excludes periodic or recurrent entropy dynamics. This conclusion does not exclude asymptotic relaxation or apply directly to a changing flow topology.

Entropy stationarity requires both a B-inflow and an F-outflow at every non-isolated vertex, together with numerical equality of the two aggregate contributions. Cycle existence alone does not establish this balance. The copying cycle demonstrates a stationary system satisfying all assumptions, while the reversible swap shows that sources, sinks, and acyclic supports are compatible with them away from stationarity. The controlled update in Eq. (3.21) demonstrates that reversibility alone does not imply strong blocking.

If no backward reconstruction set exists, the reversible framework does not apply and a residual term may remain in an entropy expansion. Local stochastic examples can still illustrate higher-order flow quantities, but they do not by themselves establish the assumptions of the main propositions. Further work could investigate how graph-theoretic properties, including hypergraph Laplacians [37], constrain information dynamics, and how explosive behavior in higher-order systems [2] relates to entropy stationarity.

A. Proof of Proposition 3.2

A.1 *A Reversible Counterexample*

Consider four binary vertices $V = \{X, C, Y, D\}$ and the deterministic reversible update

$$X' = C, \qquad C' = X, \qquad Y' = Y \oplus D, \qquad D' = D. \tag{A.1}$$

At the step under consideration, take $X = 0$, let C, D be independent fair bits, and set $Y = C \oplus N$, where $N \sim \mathrm{Bern}(\varepsilon)$ is independent of C, D and $0 < \varepsilon < 1/2$. The auxiliary variable N specifies the initial distribution and is not an additional dynamical vertex. Define

$$h(\varepsilon) := -\varepsilon \ln \varepsilon - (1-\varepsilon)\ln(1-\varepsilon), \qquad \mu := \ln 2 - h(\varepsilon) > 0. \tag{A.2}$$

Then $K_X = \{C\}$, $K_C = \varnothing$, $K_Y = \{D\}$, and $K_D = \varnothing$. The only nonzero direct TE hyperarcs are

$$\overrightarrow{E}_T = \{(\{C\},\{X\}),(\{D\},\{Y\})\}, \qquad T_{C\to X} = T_{D\to Y} = \ln 2. \tag{A.3}$$

To check that joint target sets add no further direct arcs, note that a target containing C or D contains a future variable determined by the conditioning variables. The only remaining multiple target is $\{X,Y\}$, for which $K_{\{X,Y\}} = \{C,D\}$. Conditional on Y, the variables C and D are independent, while $X' = C$ and $Y' = Y \oplus D$; all associated multiple TE terms therefore vanish. Both C and D are needed to determine this target, so replacing C by its imperfect correlate Y does not create another minimal set.

On the other hand,

$$T_{Y\to X} = I(C;Y) = \mu > 0. \tag{A.4}$$

Thus $(\{Y\},\{X\}) \in \overrightarrow{E}_{\text{total}}$, although no weak hyperpath leads from Y to X in the direct graph in Eq. (A.3). This proves Proposition 3.2. In addition,

$$T_{\{Y,C\}\to\{X\}} = \mu, \tag{A.5}$$

which explains why the neighbor-anchored support restores this connection. The example is not required to satisfy strong blocking, which is introduced only for the later entropy results. Its dynamics is nevertheless globally reversible.

The following auxiliary observations distinguish propagation through dynamical ancestors from dependence inherited through an initial distribution or a shared ancestor.

A.2 *Total Forward Neighbors*

For a set S, define $R_S^1 := \bigcup_{K\in\overline{\mathscr{K}}_S} K$ and, recursively,

$$R_S^{j+1} := \bigcup_{K\in\overline{\mathscr{K}}_{R_S^j}} K, \qquad K_S^{\text{total}} := \bigcup_{j\geqslant 1} R_S^j. \tag{A.6}$$

The union is a subset of the finite set V, so it is represented by finitely many layers. An empty layer remains empty. For the multi-step statements below, the selected neighbor relations are assumed valid throughout the time interval under consideration.

LEMMA A.1 If $u \in K_S^{\text{total}}$, then some $s \in S$ is connected to u in $\mathscr{G}_T$.

Proof. First let $u \in K_S$ for some $K_S \in \overline{\mathscr{K}}_S$, and put $W = K_S \setminus \{u\}$. Minimality gives $H_{S'}|_{S\cup W} > 0$, so some $s \in S$ satisfies $H_{s'}|_{S\cup W} > 0$. Set $L_s = (S\setminus\{s\})\cup W$. Since $H_{s'}|_{\{s,u\}\cup L_s} = 0$ but $H_{s'}|_{\{s\}\cup L_s} > 0$, reducing $L_s \cup \{u\}$ to a minimal sufficient set yields $K_s \in \overline{\mathscr{K}}_s$ containing u. Therefore

$$T_{u\to s|K_s\setminus\{u\}} := I(u;s' \mid \{s\}\cup[K_s\setminus\{u\}]) = H_{s'}|_{\{s\}\cup[K_s\setminus\{u\}]} > 0. \tag{A.7}$$

The conditioning expansion is

$$T_{u\to s|K_s\setminus\{u\}} = \sum_{W\subseteq K_s\setminus\{u\}} (-1)^{|W|} T_{\{u\}\cup W\to\{s\}}. \tag{A.8}$$

Hence at least one term on the right is nonzero. Its source contains u and is contained in K_s, so it defines a direct B-arc connecting u to s. Applying this argument successively to the layers R_S^j constructs a weak hyperpath to S. □

The set $S^{\text{total}} := S \cup K_S^{\text{total}}$ is autonomous when these neighbor relations hold throughout the interval: the future state of each layer is determined by that layer and the next, all of which belong to S^{total}.

A.3 *Two Independent Autonomous Systems*

Let S^{total} and U^{total} be disjoint autonomous vertex sets.

LEMMA A.2 Suppose $I^{\text{co}}_{A\cup B}(t_0) = 0$ for every nonempty $A \subseteq S^{\text{total}}$ and nonempty $B \subseteq U^{\text{total}}$. Then these autonomous systems remain independent at all pairs of times $t, t' \geqslant t_0$. In particular, their cross-system co-information and transfer entropy vanish.

Proof. Inclusion–exclusion expresses $I(S^{\text{total}}; U^{\text{total}})$ as a signed sum of the mixed co-informations in the hypothesis, so this block mutual information vanishes at t_0. Thus the initial joint distribution factorizes. Autonomy gives

$$\Gamma_{S^{\text{total}}}(t) = F_t(\Gamma_{S^{\text{total}}}(t_0)), \qquad \Gamma_{U^{\text{total}}}(t') = G_{t'}(\Gamma_{U^{\text{total}}}(t_0)). \tag{A.9}$$

Functions of independent initial blocks remain independent. Every mixed co-information, including the conditional quantities defining cross-system TE, therefore vanishes. □

A.4 *Indirect Statistical Relationships*

Consider $(U,S) \in \overrightarrow{E}_{\text{total}}$. If $U \cap K_S^{\text{total}} \neq \varnothing$, Lemma A.1 gives a weak hyperpath from U to S. If $U \cap K_S^{\text{total}} = \varnothing$ and the two autonomous closures are disjoint, a nonzero TE requires dependence between their initial distributions by Lemma A.2. The reversible example in Section A.1 has precisely this form. If the autonomous closures overlap, a vertex in their intersection can act as a shared dynamical ancestor. Such dependence need not yield a directed path from the observed source to the target.

B. Proof of Proposition 3.3

Since $\overrightarrow{E}_{\bar{T}} \subseteq \overrightarrow{E}_{\text{total}}$, every weak hyperpath in $\mathscr{G}_{\bar{T}}$ also exists in $\mathscr{G}_{\text{total}}$. For the reverse inclusion, consider $(U,S) \in \overrightarrow{E}_{\text{total}}$ and fix $K_S \in \overline{\mathscr{K}}_S$. Write $T_{U\to S|K_S} := I^{\text{co}}_{S'\cup U}|_{S\cup K_S}$ for the additionally conditioned multiple TE. If $U \cap K_S \neq \varnothing$, the arc already belongs to $\overrightarrow{E}_{\bar{T}}$. Otherwise, forward sufficiency and the conditioning expansion give

$$0 = T_{U\to S|K_S} = T_{U\to S} + \sum_{\varnothing\neq W\subseteq K_S} (-1)^{|W|} T_{U\cup W\to S}. \tag{B.1}$$

Since $T_{U\to S} \neq 0$, at least one nonempty W has $T_{U\cup W\to S} \neq 0$. The arc $(U\cup W, S)$ belongs to $\overrightarrow{E}_{\bar{T}}$ and connects every vertex of U to every vertex of S in the weak-path sense. Replacing each total-support arc in a weak hyperpath by such an anchored arc preserves its endpoints. Thus $\mathscr{G}^*_{\text{total}} \subseteq \mathscr{G}^*_{\bar{T}}$, proving equality.

C. Local Stochastic Illustrations of Higher-Order Channels

The following examples illustrate local flow quantities and their hyperarc orientations. The focal updates are deterministic given their displayed inputs, but the inputs are refreshed stochastically. Backward reconstruction of every vertex is not assumed, so these examples are not used as realizations of the reversible framework in Sections 4 and 5. In particular, the reversed quantities below are not asserted to form the backward-neighbor hypergraph of Eq. (3.8). The fully specified copying cycle in Section 3.3 supplies the example satisfying the main assumptions.

C.1 *A B-Arc from the XOR Operation*

Consider a focal vertex p with stochastic inputs a, b:

$$a' \sim \mathrm{Bern}(1/2), \qquad b' \sim \mathrm{Bern}(1/2), \qquad p' = a \oplus b. \tag{C.1}$$

The two fresh bits are independent of each other and of the current state. Evaluate the quantities under the invariant law in which a, b, p are independent fair bits. Then $K_p = \{a, b\}$ and

$$\begin{aligned} T_{a\to p} = T_{b\to p} = 0, \qquad & T^{\dagger}_{p\to a} = T^{\dagger}_{p\to b} = 0, \\ T_{\{a,b\}\to\{p\}} = -\ln 2, \qquad & T^{\dagger}_{\{p\}\to\{a,b\}} = -\ln 2, \\ T_{\{p\}\to\{a,b\}} = 0, \qquad & T^{\dagger}_{\{a,b\}\to\{p\}} = 0. \end{aligned} \tag{C.2}$$

Thus the displayed B-channel and its reversed counterpart have opposite orientations, as shown in Fig. C.1(a). At p, however, $\alpha_1^{K_p\to p} = -\ln 2$: its current state is not reconstructible from the displayed future variables. This is consistent with $\Delta H_p = 0$ under the invariant law and distinguishes this local illustration from the reversible propositions.

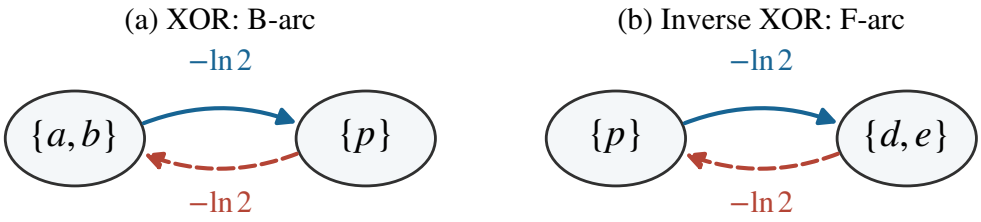


FIG. C.1. Selected higher-order channels. (a) The XOR B-arc $\{a, b\} \to \{p\}$. (b) The F-arc $\{p\} \to \{d, e\}$ in the stochastic inverse example. Solid arrows show the forward quantities and dashed arrows their reversed counterparts; each displayed quantity equals $-\ln 2$. Panel (b) highlights one channel, not the complete support, which also contains the ordinary, B-, and multiple-tail/multiple-head channels listed in Eqs. (C.4)–(C.5).

C.2 *An F-Arc from a Stochastic Inverse of XOR*

Consider stochastic inputs p, ζ and the deterministic focal update of $\{d, e\}$:

$$p' \sim \mathrm{Bern}(1/2), \qquad \zeta' \sim \mathrm{Bern}(1/2), \qquad d' = \zeta, \qquad e' = \zeta \oplus (1 - p). \tag{C.3}$$

The fresh inputs are independent of the current state and of each other. The map $(p, \zeta) \mapsto (d', e')$ is bijective and satisfies $p = 1 \oplus d' \oplus e'$. Equivalently, it is a stochastic inverse of XOR after relabeling the input as $\bar{p} = 1 - p$. Thus, for a fixed p, it chooses equally between the two output pairs with the corresponding parity. We again use the invariant law in which the four current variables are independent fair bits.

The complete nonzero TE channel list for the deterministic focal targets is

$$\begin{aligned} T_{\zeta\to d} = \ln 2, \qquad & T_{\{p,\zeta\}\to\{e\}} = -\ln 2, \\ T_{\{p\}\to\{d,e\}} = -\ln 2, \qquad & T_{\{p,\zeta\}\to\{d,e\}} = -\ln 2. \end{aligned} \tag{C.4}$$

Here $K_d = \{\zeta\}$, $K_e = \{p, \zeta\}$, and $K_{\{d,e\}} = \{p, \zeta\}$. The first channel is an ordinary edge, the second is a B-arc, the third is an F-arc, and the fourth has multiple vertices in both tail and head. Its presence is important even though the entropy analysis selects the BF substructure.

The corresponding nonzero reversed quantities are

$$\begin{aligned} T^{\dagger}_{d\to\zeta} &= \ln 2, & T^{\dagger}_{\{e\}\to\{p,\zeta\}} &= -\ln 2, \\ T^{\dagger}_{\{d,e\}\to\{p\}} &= -\ln 2, & T^{\dagger}_{\{d,e\}\to\{p,\zeta\}} &= -\ln 2. \end{aligned} \tag{C.5}$$

The pairwise quantities $T_{p\to d}$, $T_{p\to e}$, $T_{\zeta\to e}$, $T^{\dagger}_{d\to p}$, and $T^{\dagger}_{e\to p}$ vanish. The four listed channels therefore exhibit a reversal of support at the level of these local quantities. This observation is not a claim that the refreshed-input system satisfies the global reversible assumptions or possesses all backward-neighbor sets. Figure C.1(b) highlights the F-arc; the complete list is given in Eqs. (C.4)–(C.5).

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